

Creating a **Point-to-Site** (P2S) VPN in **Microsoft Azure**

A Point-to-Site VPN connection lets securely connect individual clients (like laptops or mobile devices) to Azure Virtual Network (VNet) without a physical or direct network connection. This can be useful for remote workers or developers needing secure access to resources in a VNet.

Prerequisites

1. **Azure Subscription:** Need an Azure account.
2. **Virtual Network (VNet):** Create a VNet, if don't have one already. Ensure it has a subnet for the VPN Gateway.
3. **Subnet for VPN Gateway:** A dedicated subnet is required for the gateway. It's usually named **GatewaySubnet**.

Flowchart Diagram Example

1. Create Virtual Network (VNet)



2. Deploy Virtual Network Gateway (VNG)



3. Configure VPN Gateway for P2S Connections

4. Generate/Upload Root Certificate for Authentication

5. Configure Client VPN Software

6. Connect Client to VPN

7. Verify Connection

Steps to Create a Point-to-Site VPN in Azure

Step 1: Create a Virtual Network (VNet) in Azure

1. **Log into Azure Portal** (portal.azure.com).
2. In the left-hand menu, click **Create a resource**.
3. Search for **Virtual Network** and select it.
4. Click **Create**.
5. In the **Create Virtual Network** pane:
 - **Subscription**: Choose subscription.
 - **Resource group**: Choose an existing resource group or create a new one.
 - **Name**: Provide a name for the VNet (e.g., AzureVNet).
 - **Region**: Choose a region close to on-premises network.

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Azure Virtual Network (VNet) is the fundamental building block for your private network in Azure. VNet enables many types of Azure resources, such as Azure Virtual Machines (VM), to securely communicate with each other, the internet, and on-premises networks. VNet is similar to a traditional network that you'd operate in your own data center, but brings with it additional benefits of Azure's infrastructure such as scale, availability, and isolation. [Learn more.](#)

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription * Azure subscription 1

Resource group * (New) RG [Create new](#)

Instance details

Virtual network name * VNET

Region * (Asia Pacific) Central India [Deploy to an Azure Extended Zone](#)

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- **IP Addressing:** Define the address space (e.g., 10.0.0.0/16).
- **Subnets:** Create a subnet (e.g., 10.0.1.0/24).

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Configure your virtual network address space with the IPv4 and IPv6 addresses and subnets you need. [Learn more](#)

Define the address space of your virtual network with one or more IPv4 or IPv6 address ranges. Create subnets to segment the virtual network address space into smaller ranges for use by your applications. When you deploy resources into a subnet, Azure assigns the resource an IP address from the subnet. [Learn more](#)

+ Add a subnet

10.0.0.0/16

Delete address space

10.0.0.0 - 10.0.255.255
65,536 addresses

Subnets	IP address range	Size	NAT gateway
VNETSUB	10.0.1.0 - 10.0.1.255	/24 (256 addresses)	-

Add IPv4 address space

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6. After reviewing the configuration, click **Create**.

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Basics

Subscription	Azure subscription 1
Resource Group	RG
Name	VNET
Region	Central India

Security

Azure Bastion	Disabled
Azure Firewall	Disabled
Azure DDoS Network Protection	Disabled

IP addresses

Address space	10.0.0.0/16 (65,536 addresses)
Subnet	VNETSUB (10.0.1.0/24) (256 addresses)

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Step 2: Create a Virtual Network Gateway

1. In the Azure portal, search for **Virtual network gateways**.
2. Click **Add** to create a new gateway.
3. In the **Create virtual network gateway** pane:
 - **Subscription:** Select the subscription where VNet is located.
 - **Resource Group:** Select the same resource group as VNet.
 - **Region:** Select the same region as VNet.
 - **Name:** Provide a name for the gateway (e.g., AzureVNetGateway).
 - **Gateway Type:** Select **VPN**.
 - **VPN Type:** Choose **Route-based** (recommended for most scenarios).
 - **SKU:** Select a SKU (e.g., **VpnGw1** or **VpnGw2** depending on needs).
 - **Virtual Network:** Select the VNet created earlier.

- **Public IP Address:** Select **Create new** and give it a name (e.g., AzureVPNPublicIP).
- **Configure BGP:** Choose **Disabled** unless you are using BGP for routing.

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Subscription: [vpngw1](#)

Generation: [Generation1](#)

Virtual network: [\(New\) VNET](#)
[Create virtual network](#)

Only virtual networks in the currently selected subscription and region are listed.

Gateway subnet address range: [10.0.1.0/24](#)
 10.0.1.0 - 10.0.1.255 (256 addresses)

Public IP address

Public IP address: ☒ Create new ☐ Use existing

Public IP address name: [AzureVPNPublicIP](#)

Public IP address SKU: Standard

Assignment: ☐ Dynamic ☒ Static

Enable active-active mode: ☐ Enabled ☒ Disabled

Configure BGP: ☐ Enabled ☒ Disabled

Authentication Information (Preview)

Enable Key Vault Access: ☐ Enabled ☒ Disabled

Azure recommends using a validated VPN device with your virtual network gateway. To view a list of validated devices and instructions for configuration, refer to Azure's [documentation](#) regarding validated VPN devices.

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4. Click **Review + Create**, and after validation, click **Create**.

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Basics

Subscription	Azure subscription 1
Resource group	RG
Name	AzureVNetGateway
Region	Central India
SKU	VpnGw1
Generation	Generation1
Virtual network	VNET
Subnet	GatewaySubnet (10.0.1.0/24)
Gateway type	Vpn
VPN type	RouteBased
Enable active-active mode	Disabled
Configure BGP	Disabled
Public IP address	AzureVPNPublicIP

Tags

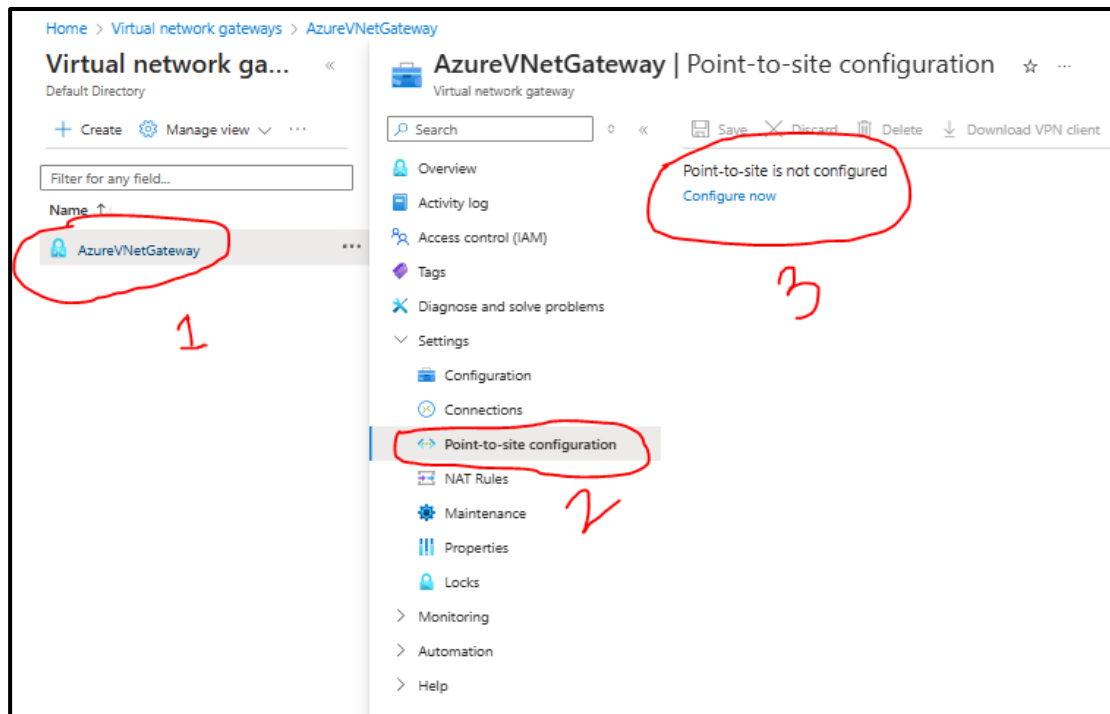
None

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Note: The Virtual Network Gateway takes 30 minutes or more to deploy, depending on the SKU.

Step 3: Configure Point-to-Site VPN Settings

1. Go to the **Virtual Network Gateway** created.
2. In the left pane, select **Point-to-site configuration**.



3. Configure the P2S VPN settings:

- **Address Pool:** Define an IP address range for VPN clients (e.g., 172.16.201.0/24). This range should not overlap with VNet address space.
- **Tunnel Type:** Choose **IKEv2** and **SSTP (SSL)** if want compatibility with Windows and Mac.
- **Authentication Type:** Use **Certificate**-based authentication:
 - For **Certificates:** Generate a root certificate and client certificates for users (need to upload the root certificate in this case).

4. Root Certificate Creation and Uploading

- Open PowerShell and past the below commend



Root_Certificate.txt

```
PowerShell Copy

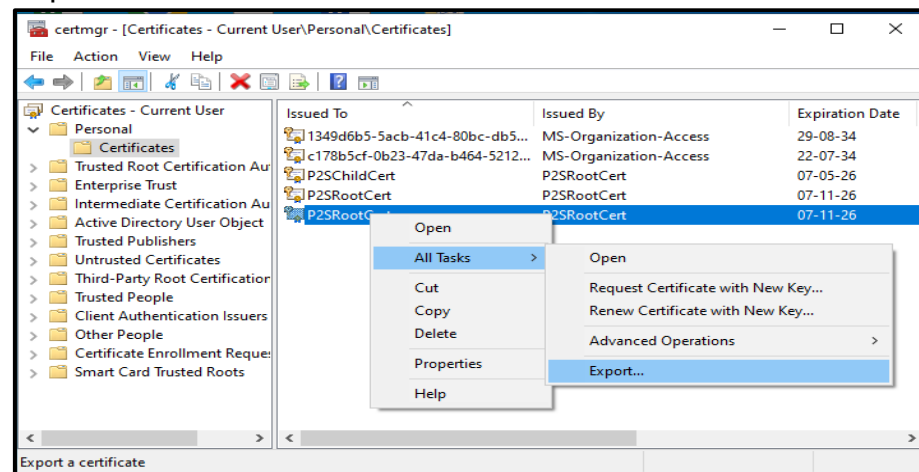
$params = @{
    Type = 'Custom'
    Subject = 'CN=P2SChildCert'
    DnsName = 'P2SChildCert'
    KeySpec = 'Signature'
    KeyExportPolicy = 'Exportable'
    KeyLength = 2048
    HashAlgorithm = 'sha256'
    NotAfter = (Get-Date).AddMonths(18)
    CertStoreLocation = 'Cert:\CurrentUser\My'
    Signer = $cert
    TextExtension = @(
        '2.5.29.37={text}1.3.6.1.5.5.7.3.2')
}
New-SelfSignedCertificate @params
```

```
PS C:\Windows\system32>
PS C:\Windows\system32>
PS C:\Windows\system32>
PS C:\Windows\system32> $params = @{
>> Type = 'Custom'
>> Subject = 'CN=P2SRootCert'
>> KeySpec = 'Signature'
>> KeyExportPolicy = 'Exportable'
>> KeyUsage = 'CertSign'
>> KeyUsageProperty = 'Sign'
>> KeyLength = 2048
>> HashAlgorithm = 'sha256'
>> NotAfter = (Get-Date).AddMonths(24)
>> CertStoreLocation = 'Cert:\CurrentUser\My'
>> }
PS C:\Windows\system32> $cert = New-SelfSignedCertificate @params
PS C:\Windows\system32>
```

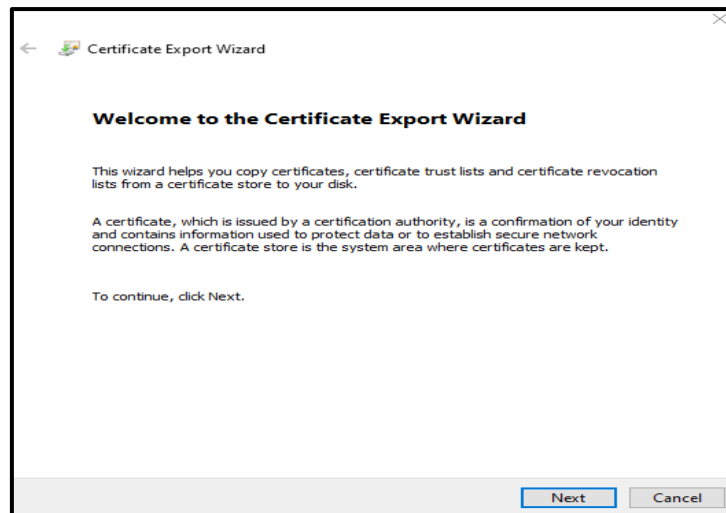
- Open Certificate console and verify the root certificate entry

Note: - don't close PowerShell console during this activity.

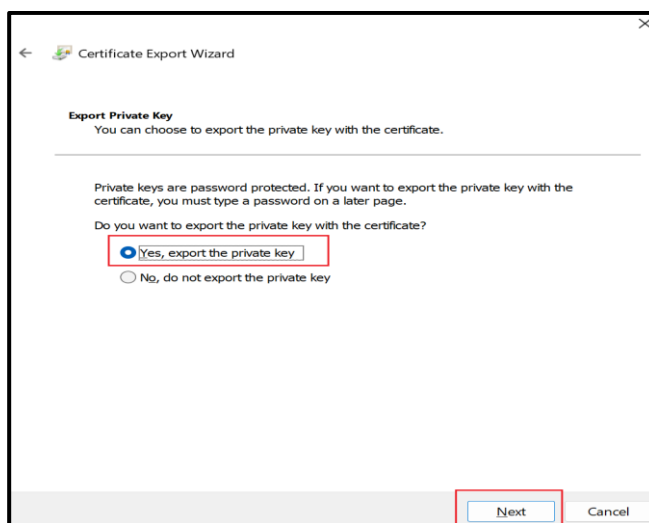
- Export the root certificate from certificate console



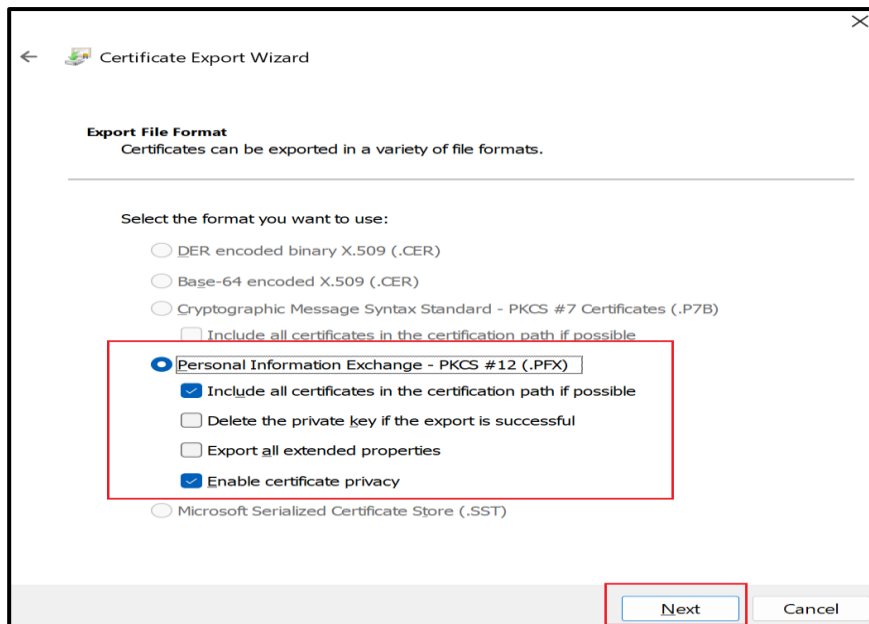
- Then open certificate wizard and click on **Next**



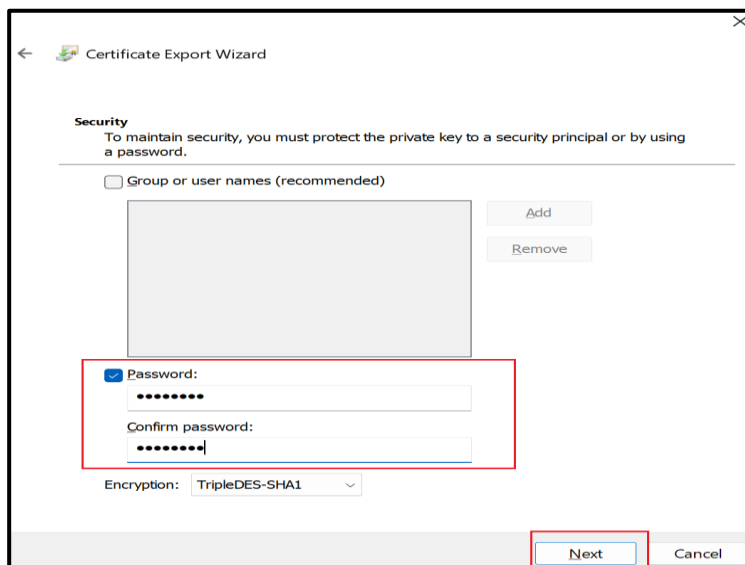
- In the Certificate Export Wizard, click **Next** to continue.
- Select **Yes, export the private key**, and then click **Next**.



- On the **Export File Format** page, leave the defaults selected. Make sure that **Include all certificates in the certification path if possible** is selected. This setting additionally exports the root certificate information that is required for successful client authentication. Without it, client authentication fails because the client doesn't have the trusted root certificate. Then, click **Next**.



- On the **Security** page, you must protect the private key. If you select to use a password, make sure to record or remember the password that you set for this certificate. Then, click **Next**.



- On the **File to Export**, **Browse** to the location to which you want to export the certificate. For **File name**, name the certificate file. Then, click **Next**.
- Click **Finish** to export the certificate.

Step 3: Download the VPN Client

1. Once the P2S configuration is saved, go back to the **Point-to-site configuration** pane.

2. Select **Download VPN client**. This downloads a pre-configured VPN client package for Windows or other devices.

Step 4: Install and Connect the VPN Client

1. **Install the VPN client** on your client device.
2. Open the VPN client and select **Connect** to initiate the connection.
3. If using certificate-based authentication, ensure that client certificates are installed on the device.